

Candidate Name: _____

Class Adm No

--	--



Promotional Examination II 2011

Pre-university 2

Biology Higher 1

Paper 2
Syllabus 8875/02

Tuesday

13 September 2011

2 hours

Additional materials:
Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, Adm No. and class on all the papers you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on question booklet.

Section B

Answer any **one** question on writing paper.

At the end of the examination, fasten all your work securely together.

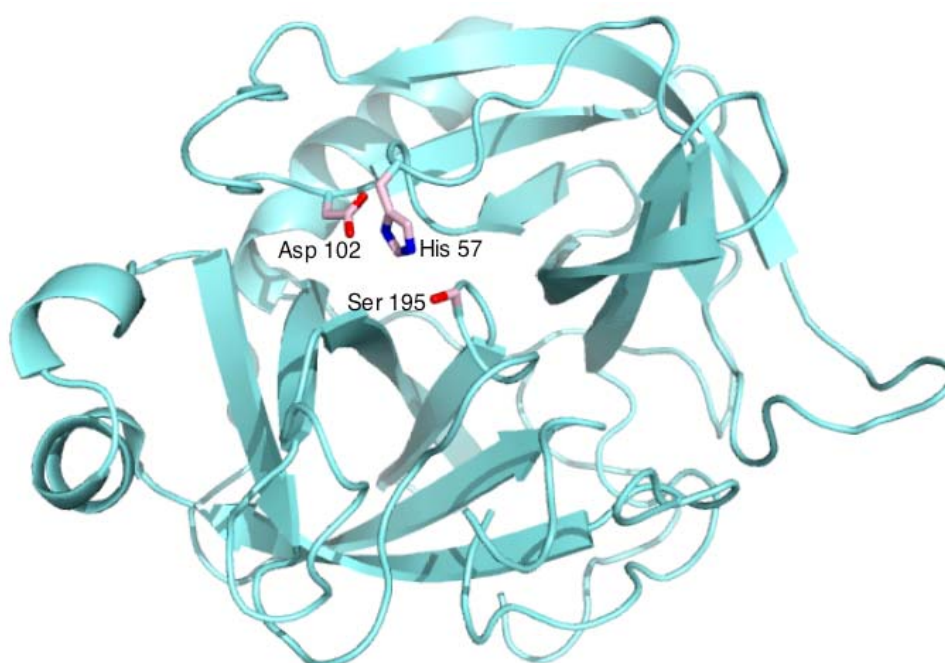
The number of marks is given in brackets[] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5 or 6	
Total	

Section AAnswer **all** questions in this section.

1. Chymotrypsin catalyses the hydrolysis of peptide bonds of proteins. It is made up of 241 amino acids. **Fig 1.1** shows the arrangement of three amino acids being in different positions in the amino acid sequence. The active site of chymotrypsin is located in a slight depression on one side of the molecule. Three amino acids form the active site. Even though these three amino acids are some distance apart on the polypeptide chain but close together in the active site.

1	Histidine 57	Aspartic acid 102	Serine 195	241

Fig 1.1**Fig 1.2**

- (a) Describe how amino acid residues at different positions in the protein may be brought together in the active site when the enzyme is synthesised.

.....

.....

.....

.....[2]

Fig. 1.3 shows the effects of increasing substrate concentration on the rate of a chymotrypsin-catalysed reaction at a temperature of 35°C and a constant pH.

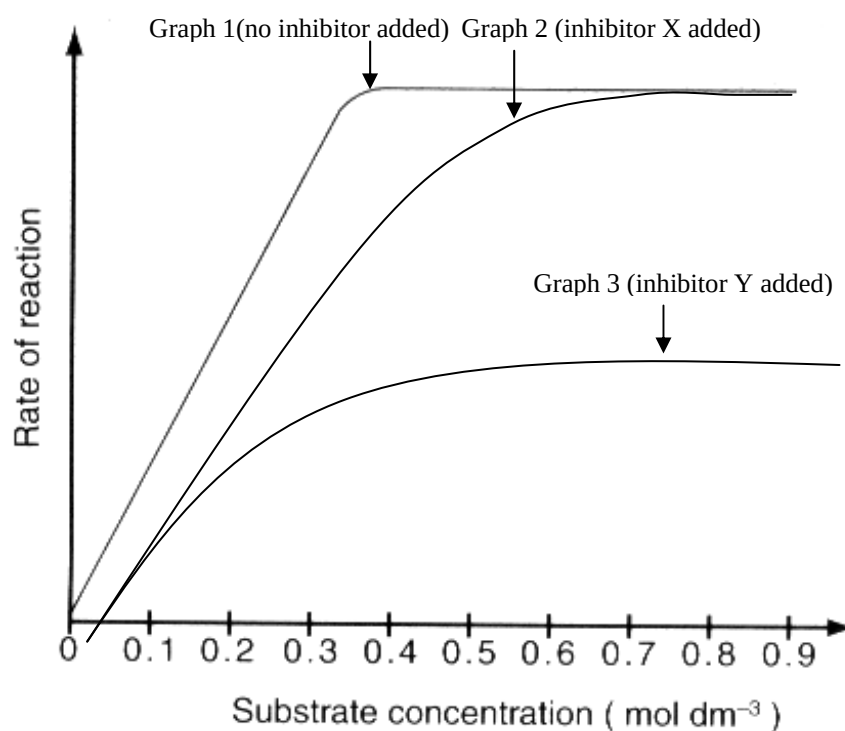


Fig 1.3

- (b) With reference to graph 1 in Fig 1.3, describe and explain the effect of substrate concentration on the rate of enzyme-catalysed reaction at high substrate concentration.

.....

[2]

- (c) Two different types of inhibitors, X and Y, were added and their results were shown in **graph 2** and **3** respectively. Describe **two** differences between inhibitor X and Y.

.....

[2]

Table 1.4 shows some mRNA codons and the corresponding amino acids for which they code. **Fig. 1.5** shows

- the sequence of 4 amino acids found in the middle section of the human chymotrypsin polypeptide
- the corresponding nucleotide bases in the DNA.

amino acid	abbreviation	mRNA codons					
glutamic acid	glu	GAA	GAG	–	–	–	–
phenylalanine	phe	UUU	UUC	–	–	–	–
lysine	lys	AAA	AAG	–	–	–	–
proline	pro	CCA	CCC	CCG	CCU	–	–
threonine	thr	ACA	ACC	ACG	ACU	–	–
valine	val	GUA	GUC	GUG	GUU	–	–
cysteine	cys	UGC	UGU	–	–	–	–
arginine	arg	CGC	CGA	CGU	CGG	AGA	AGG

Table 1.4

Amino acid Sequence	Arg	Cys	Lys	Glu
mRNA sequence				
Coding DNA sequence	GCA	ACA	TTT	CTT

Fig. 1.5

(d)

- (i) Use the information in **Table 1.4** to complete the nucleotide sequences for the mRNA shown in **Fig. 1.5**. Write your answer on **Fig. 1.5**. **[1]**

- (ii) A mutation arises and results in a change in the nucleotide sequence shown in **Fig. 1.5** to **GCAACAATTCTT**. Identify the type of mutation that has occurred.

.....**[1]**

(iii) Describe the effect of the mutation in (d)(ii).

.....

.....

.....

.....[2]

[Total:10]

2. **Fig. 2.1** shows two organelles that are found in a liver cell.

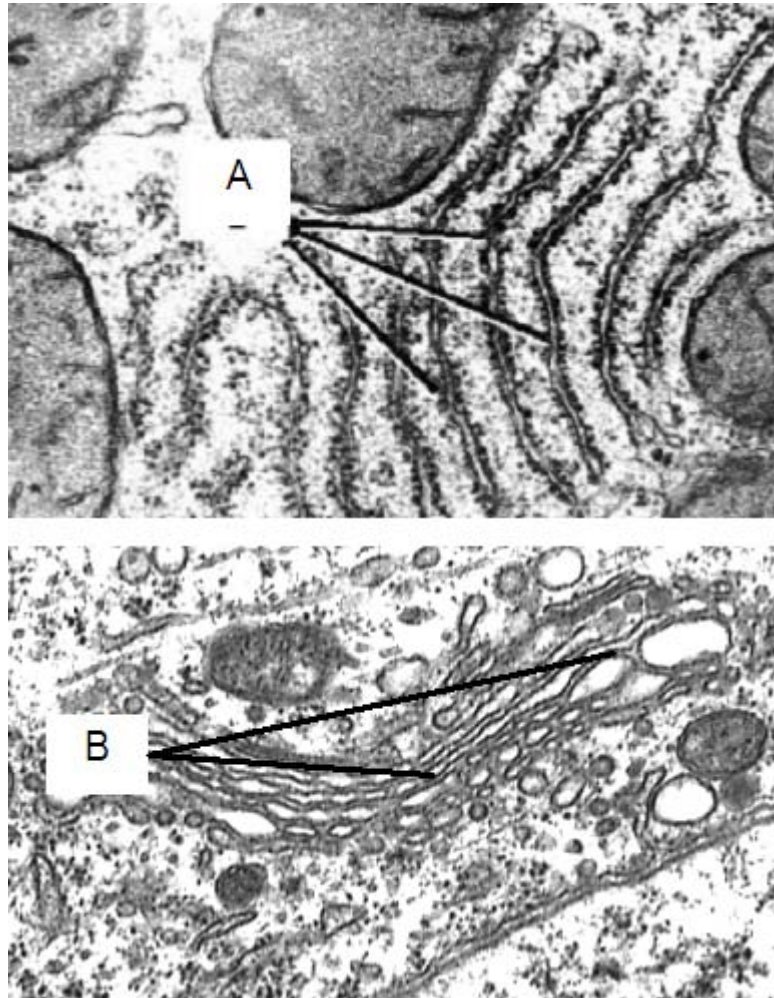


Fig. 2.1

- (a) Identify A and B in **Fig. 2.1**.

A:

B:[1]

- (b) Describe the function of **A** and **B** with relation to the synthesis of a protein found outside the cell.

.....

.....

.....

.....

.....

.....[3]

Fig 2.2 shows a diagram of a ribosome carrying out protein translation.

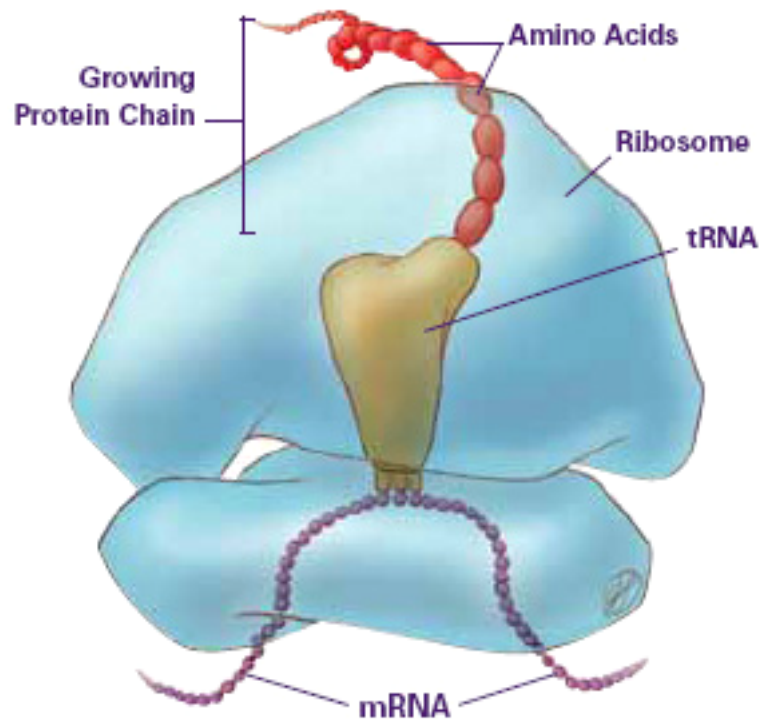


Fig. 2.2

(c)

(i) Explain how tRNA binds to mRNA.

.....

.....

.....

.....

.....[2]

(ii) Describe how the structure of the ribosome is adapted for translation.

.....

.....

.....

.....

.....[2]

Fig. 2.3 shows two diagrammatically chromosome from two cells from the cell organism undergoing different types of nuclear division.

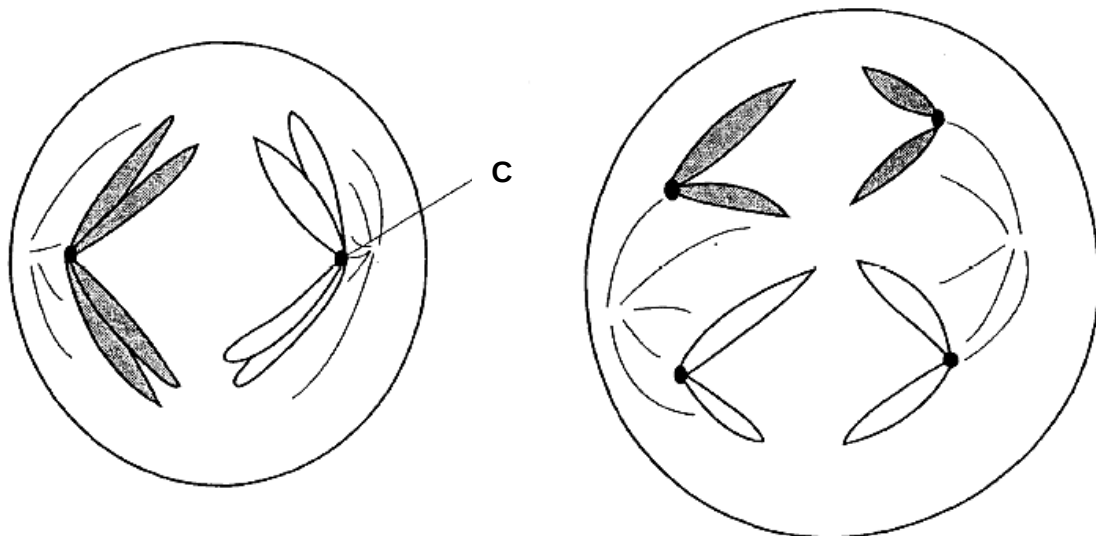


Fig. 2.3

(d) State one function of structure **C**.

.....
[1]

[Total: 9]

3. Two heterozygous pea plants were grown from round, yellow seeds. These plants were crossed and the seeds produced were recorded.

325 round, yellow

100 wrinkled, yellow

90 round, green

30 wrinkled, green

The expected phenotypic ratio for this cross is 9:3:3:1

- (a)** Explain why the observed phenotypic ratio is not 9:3:3:1.

.....
.....[1]

- (b)** Using suitable symbols, construct a genetic diagram to explain the results.

[4]

Evolutionary changes in the beaks of Darwin's finches have been instrumental in the adaptive radiation of these birds in the Galapagos Islands. There are a dozen species that are closely related to one another showing remarkable differences in beak shape and size, and in eating habits. They have all arisen from a common ancestor, but have evolved to exploit different food sources. The precise dimensions (length depth and width) of each species' beak are crucial to their survival, and fluctuations in the environment lead to selection that changes the relative success of birds with various beak shapes. The molecular basis for variation in beak size and shape appears to be due to two genes: the gene for **bone morphogenetic protein 4** (*BMP4*) that encodes a cell-cell signaling protein that is expressed more during the embryonic development of the deep and wide beaks of ground finches than during the development of narrower beaks. the gene for **calmodulin** (*CaM*) that encodes a calcium signaling protein that is expressed at much higher levels in the long and pointed beaks of cactus finch embryos.

Fig. 3.1 shows how these two genes interact to produce variation in beak size and shape.

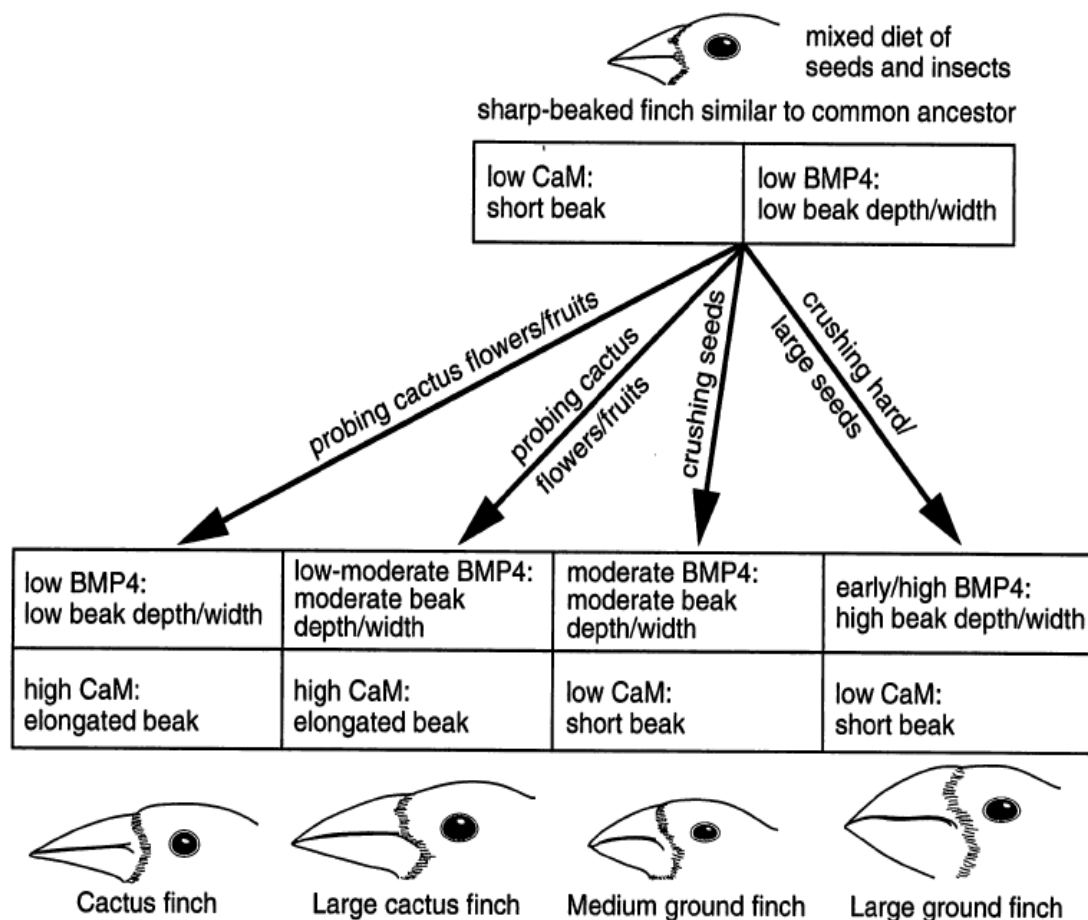


Fig. 3.1

- (c) Describe two environmental factors that drive the natural selection of finches in the Galapagos Island.

.....
.....
.....
.....[2]

- (d) Explain how the molecular evidence described supports Darwin's theory of natural selection.

.....
.....
.....
.....
.....
.....[3]

[Total: 10]

4. The polymerase chain reaction (PCR) is a method of obtaining large amounts of DNA from a small initial sample. **Fig. 4.1** shows the main stages in the polymerase chain reaction.

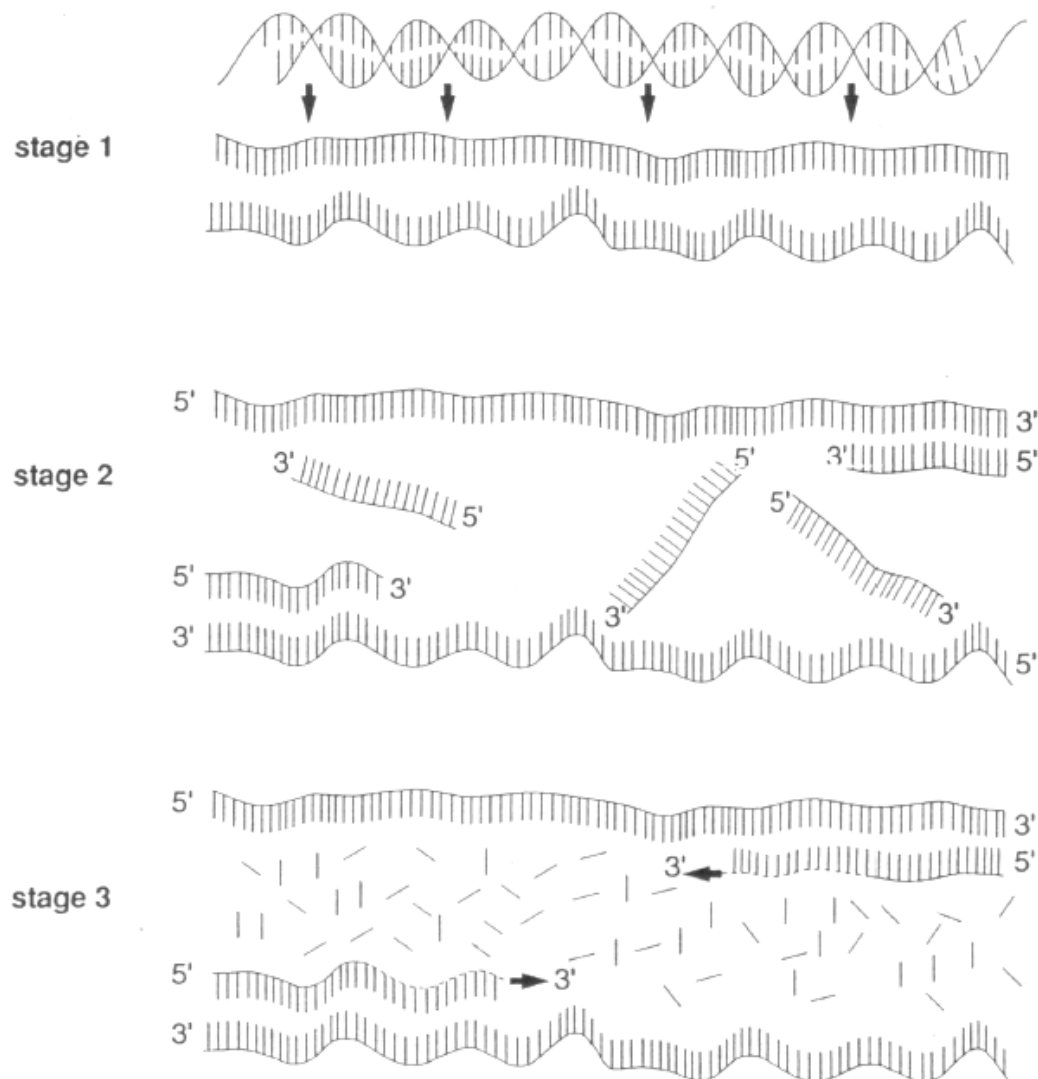


Fig. 4.1

(a)

- (i)** Explain why DNA synthesis takes place in opposite directions on the two separated strands of the double helix.

.....

.....

.....

.....

.....[2]

- (ii) With reference to **Fig. 4.1**, briefly describe what is happening at each of the three stages.

Stage 1:.....

.....[1]

Stage 2:.....

.....[1]

Stage 3:.....

.....[1]

Blood samples from 3 male adults, 1 female adult, and the baby were also taken and DNA extracted is amplified using PCR. Following PCR, the amplified DNA is cut into fragments with restriction enzymes. One of the loci on chromosome 7, **D7S820**, has many alleles, which differ from each other by the number of restriction sites present.

The amplified samples were loaded onto a gel and electrophoresis was performed to separate the fragments of DNA. They are separated by gel electrophoresis. The DNA is then treated with alkali and transferred onto a nitrocellulose membrane before radioactively labeled probes are added. The membrane is then washed and placed below a photographic film. The genetic fingerprint can be seen when the film is developed. A diagram of it is shown in **Fig. 4.3**

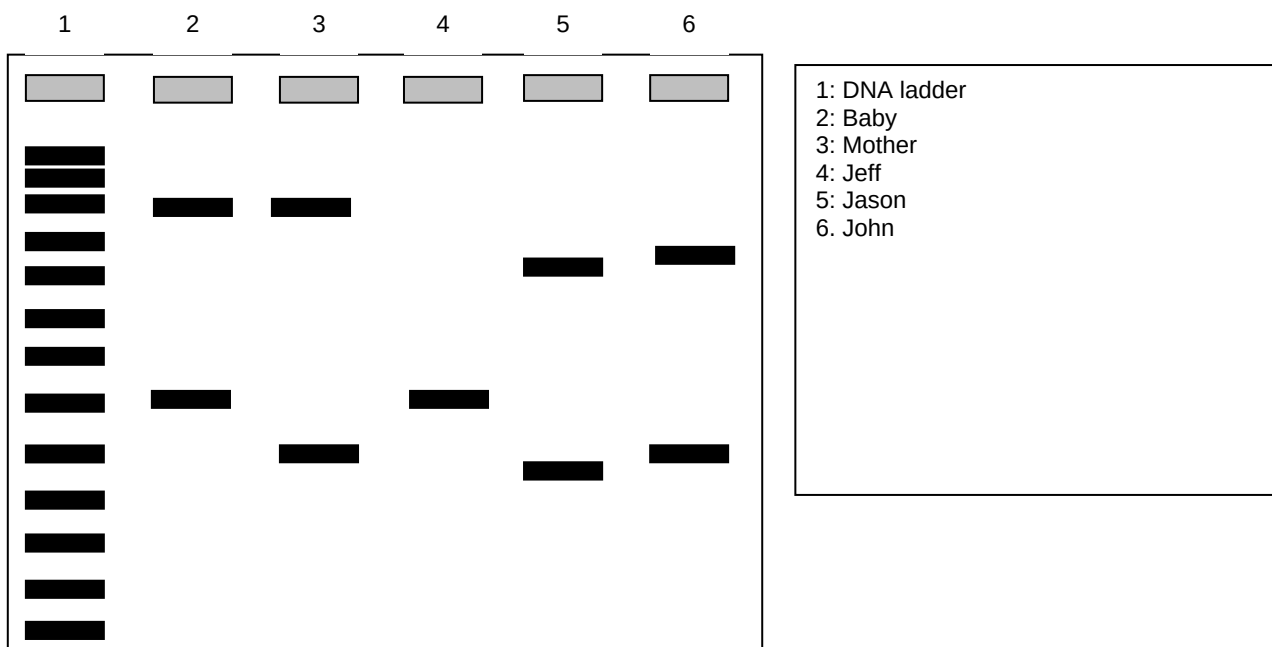


Fig. 4.3

(c)

(i) Account for the different position of the bands for each individual.

.....

.....

.....

.....

.....[2]

(ii) Who is most likely to have fathered the child?

.....

.....[1]

[Total: 11]

Section BAnswer **one** question

Write your answer to the question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections (a), (b) etc, as indicated in the question.

Either

5.

- (a) Describe the different types of stem cells and explain their unique features and normal functions in a living organism. [10]
- (b) Describe the structure and organisation of DNA in eukaryotes. [6]
- (c) Discuss how the proteins present in the membrane structure enables the membrane to be selectively permeable. [4]

[Total: 20]**Or**

6.

- (a) Explain how the structure of chloroplast is adapted to its role in carrying out photosynthesis efficiently. [8]
- (b) Describe the process of oxidative phosphorylation in respiration. [8]
- (c) Distinguish between Krebs cycle and Calvin cycle [4]

[Total: 20]**End of Paper**

